



Leibniz
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Stochastic Finite Element Methods

#2: Stochastic Computations with FEM

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The Problem at Hand

We want to solve a mechanical problem, e.g. $\operatorname{div} \boldsymbol{\sigma} + \varrho \mathbf{b} = \mathbf{0}$
within a spatial domain (body) $\mathcal{D} \in \mathbb{R}^3$

with constitutive equations $\boldsymbol{\sigma} = \mathbb{C}(x, t) \cdot \cdot (\boldsymbol{\varepsilon}^e(x, t) - \boldsymbol{\varepsilon}^i(x, t))$

where $\boldsymbol{\varepsilon} = \operatorname{sym}(\operatorname{grad} \mathbf{u})$

and $\dot{\boldsymbol{\varepsilon}}^i = \mathcal{H}(\boldsymbol{\sigma}, \mathbb{A}, t)$

subjected to boundary conditions $\mathbf{u} = \bar{\mathbf{u}} \quad \text{at} \quad \partial_u(\mathcal{S})$
 $\boldsymbol{\sigma} \cdot \mathbf{n} = \bar{\mathbf{t}} \quad \text{at} \quad \partial_t(\mathcal{S})$

Now we assume that the constitutive parameters and the loading are uncertain, i.e.

$$\mathbb{C}(x, t) \longrightarrow \mathbb{C}(x, \theta, t)$$

$$\mathbb{A}(x) \longrightarrow \mathbb{A}(x, \theta)$$

$$\bar{\mathbf{t}} \longrightarrow \bar{\mathbf{t}}(\theta)$$

The Problem at Hand

Elliptic problem of the kind

$$\operatorname{div}[\mathbb{C}(\boldsymbol{x}, \theta, t) \cdot \cdot \operatorname{sym}(\operatorname{grad} \boldsymbol{u}(\boldsymbol{x}, \theta, t))] + \varrho \boldsymbol{b}(\boldsymbol{x}, \theta) = \mathbf{0}$$

random loading

$$\boldsymbol{\sigma} \cdot \boldsymbol{n} = \bar{t}(\theta) \quad \text{at} \quad \partial_t(\mathcal{S})$$

random evolution of internal variables

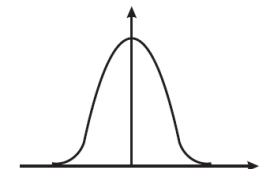
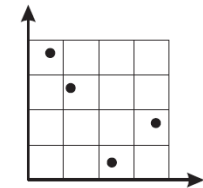
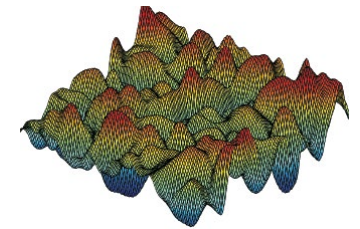
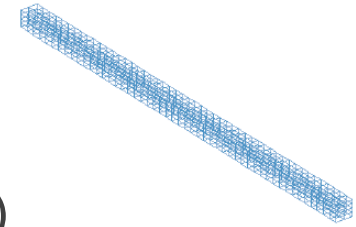
$$\dot{\boldsymbol{\varepsilon}}^i = \mathcal{H}(\boldsymbol{\sigma}(\boldsymbol{x}, \theta, t), \mathbb{A}(\boldsymbol{x}, \theta), t)$$

Finite Element Approximation

$$\mathbf{K}(\mathbf{x}, \theta) \mathbf{u}(\mathbf{x}, \theta) = \mathbf{f}(\theta) \quad \text{with} \quad \theta \in \Theta$$

Probabilistic FE-Analysis

- Structural Mechanics Model (FEM)
- Assignment of uncertain material properties (Random field modeling)
- Assignment of random loads
- Stochastic analysis
 - Sampling techniques (Monte Carlo etc.)
 - Collocation methods (Response Surfaces)
 - Perturbation / Expansion Methods
 - Spectral Stochastic Finite Element Approach
- Statistical analysis

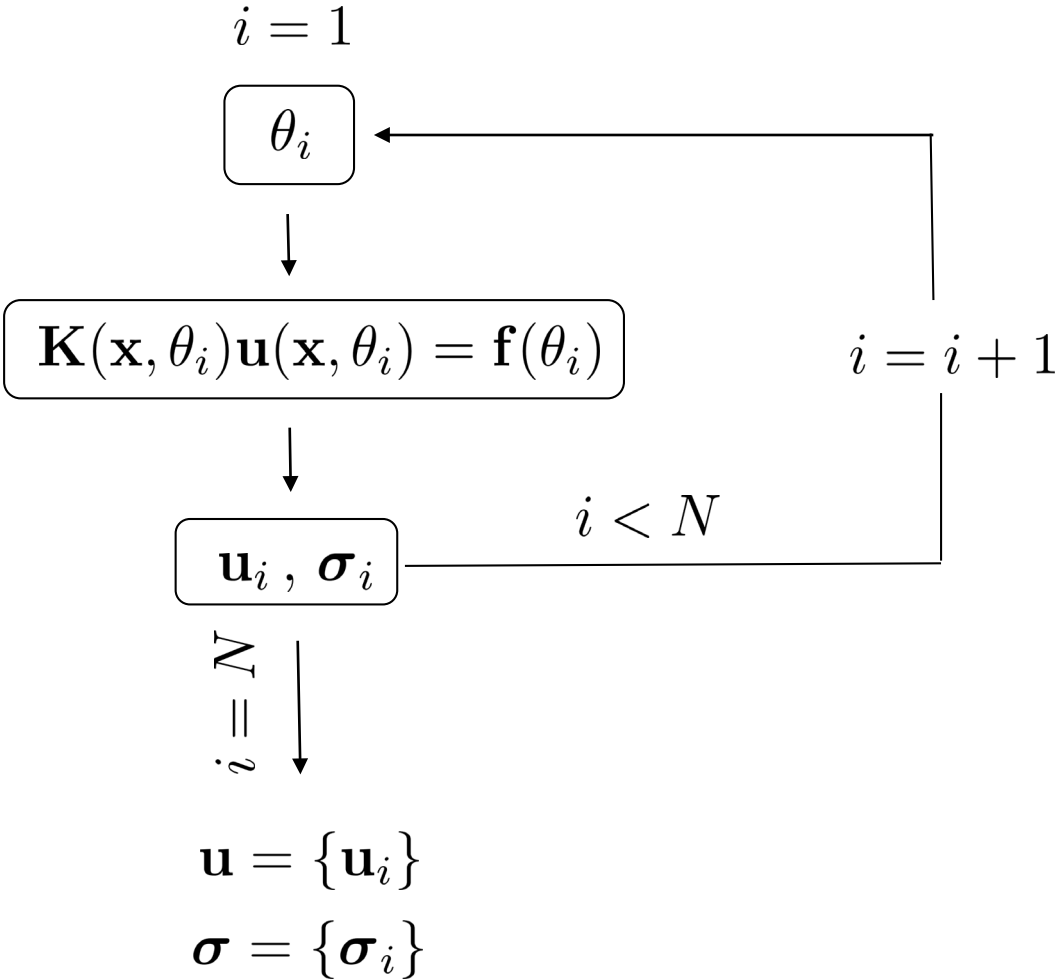


Sampling Methods

random number
generator

deterministic
solver

stochastic output



Monte Carlo Method

- Chose a (pseudo) random number generation, e.g. *rand*, *randn*
- Discretize the random input space by N random numbers
- Solve the deterministic problem
- Statistical analysis of the output space

$$\mu = \frac{1}{n} \sum_{i=1}^n y_i \quad \sigma^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \mu)^2$$

Properties:

- Independent from the stochastic dimension
- Poor convergence $\mathcal{O}(1/\sqrt{n})$

Quasi Monte Carlo Methods

- Sobol Sequences

Expansion of integers on base 2

$$x_i = \frac{b_0}{2} + \frac{b_1}{2^2} + \cdots \frac{b_m}{2^{m+1}}$$

Integers $b_k \in [0, 1]$ constructed such that

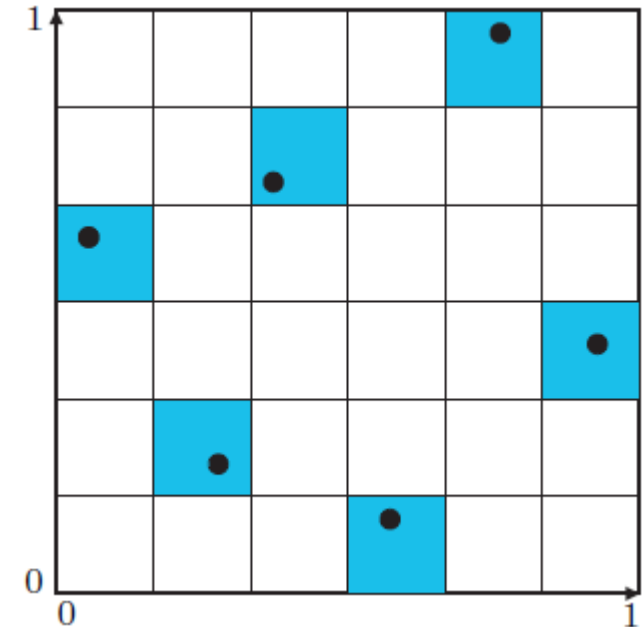
$$(i - 1)_2 = b_m b_{m-1} \cdots b_1 b_0$$

Theoretical (best) convergence rate $\mathcal{O}(\log^M(n)/n)$

M : stochastic dimension

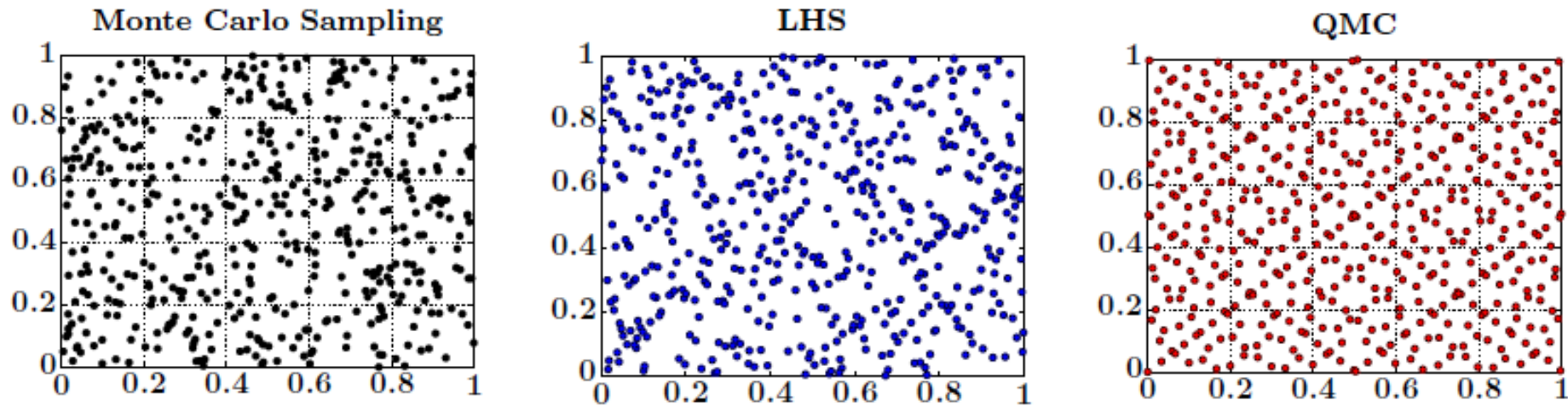
Latin Hypercube Sampling

- discretization of the random space
- approximated convergence rate $\mathcal{O}(1/n)$



QMC Methods

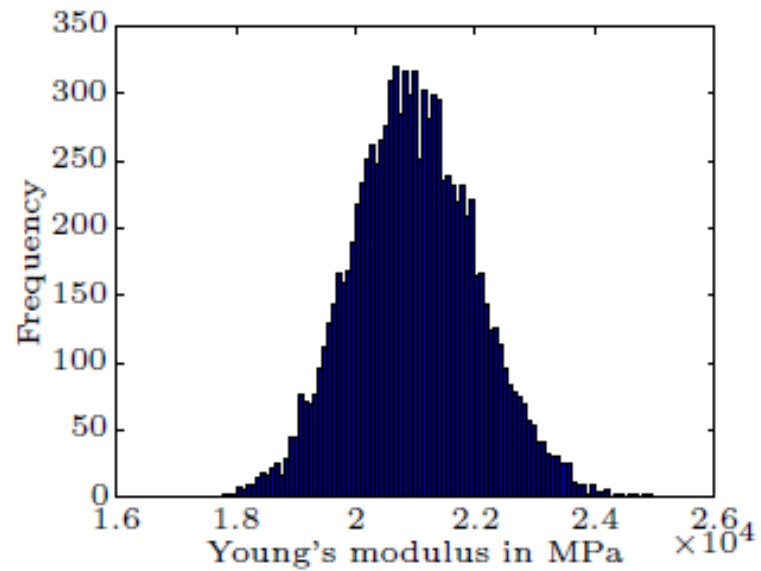
Space filling of sample points (n=512)



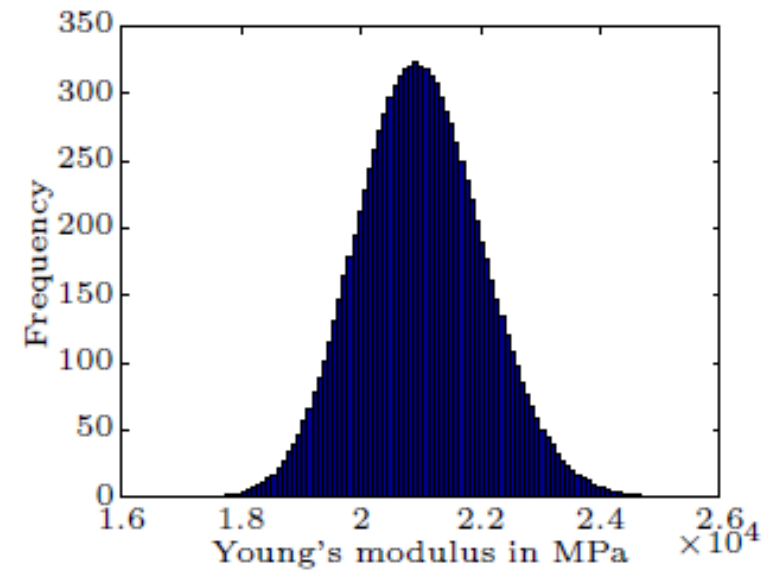
Convergence behaviour depends on the stochastic dimension!

Example

Random Youngs Modulus: $\mu_E = 210\text{GPa}$ $\sigma_E = 5\%$



Monte Carlo Sampling



Latin Hypercube Sampling

10.000 samples each

Stochastic Collocation Methods

Basic Idea: Approximation of the Random Process by a (Lagrangian) polynomial fit of discrete random response (response surface).

$$u(\mathbf{X}) = \mathcal{I}(u)(\mathbf{X}) = \sum_{k=1}^{n_{cp}} u(\mathbf{X}_k) L_k(\mathbf{X})$$

Statistics

$$\mathbb{E}\{u\} = \sum_{k=1}^{n_{cp}} u(\mathbf{X}_k) \int_{\Gamma} L_k(\mathbf{X}) f_X(\mathbf{X}) d\mathbf{X}$$

$$\text{Var}\{u\} = \sum_{k=1}^{n_{cp}} [u(\mathbf{X}_k, \mathbf{x}) - \mathbb{E}(u(\mathbf{x}))]^2 \int_{\Gamma} L_k(\mathbf{X}) f_X(\mathbf{X}) d\mathbf{X}$$

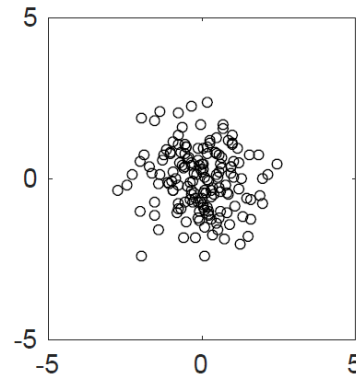
In general tensorial representation of the random space (curse of dimension):

- Full-Tensor Product
- Sparse Grids
- Smolyak Algorithm

Stochastic Collocation Methods

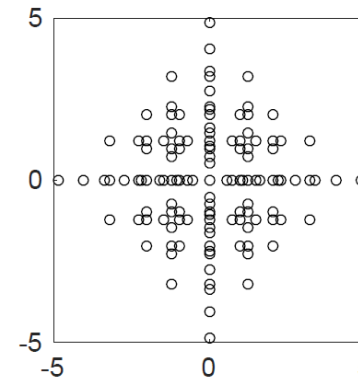
Monte Carlo simulation

- Random numbers X



Stochastic collocation

- Collocation points Y



Repetitive deterministic simulations

- Evaluation by statistical methods
- Polynomial interpolation

Full Tensor Product

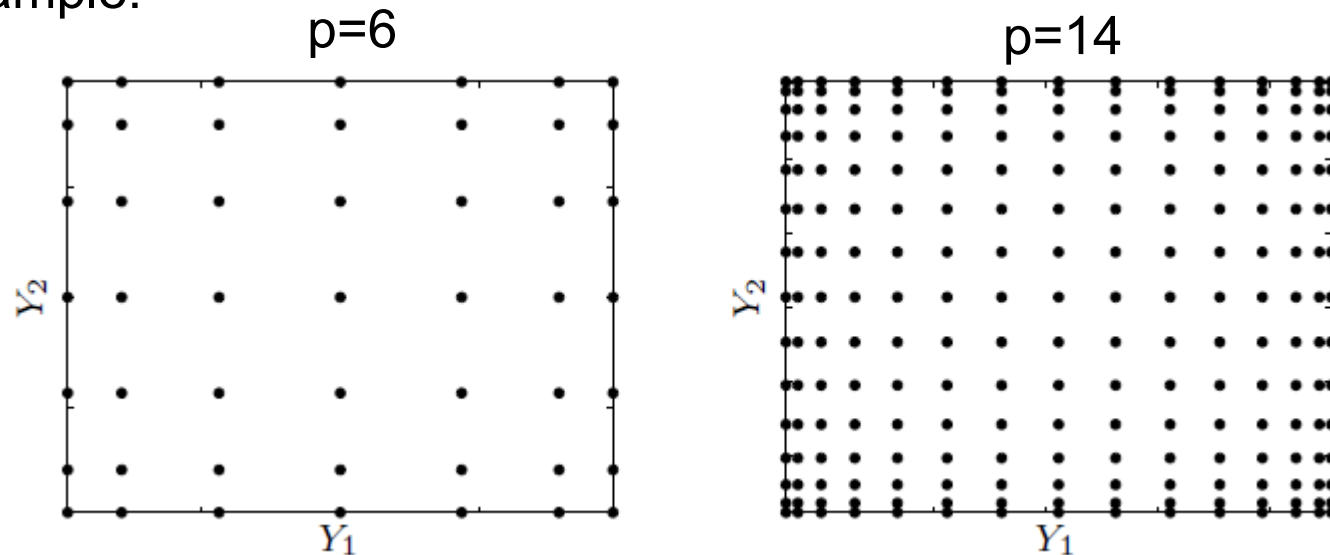
One dimensional interpolant $\mathcal{U}^i(u) \sum_{k=1}^{m_i} u(X_k^i) L_k^i$

Tensorial product space $\Theta_N = \Theta^{i_1} \otimes \dots \otimes \Theta^{i_N}$

Multi-dimensional interpolant

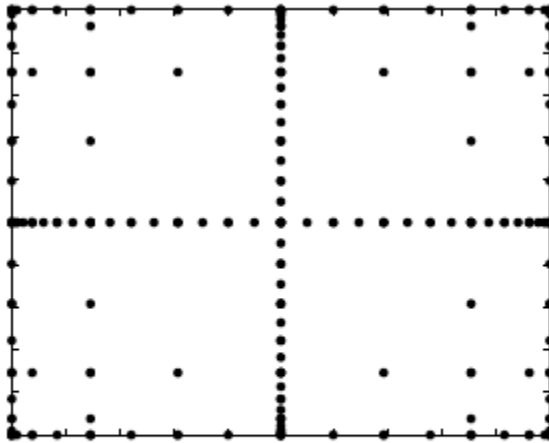
$$\begin{aligned} \mathcal{I}(u) &= (\mathcal{U}^{i_1} \otimes \dots \otimes \mathcal{U}^{i_N})(u) \\ &= \sum_{k_1=1}^{m_{i_1}} \dots \sum_{k_N=1}^{m_{i_N}} u(X_{k_1}^{i_1}, \dots, X_{k_N}^{i_N}) \cdot (L_{k_1}^{i_1} \otimes \dots \otimes L_{k_N}^{i_N}) \end{aligned}$$

2D Example:

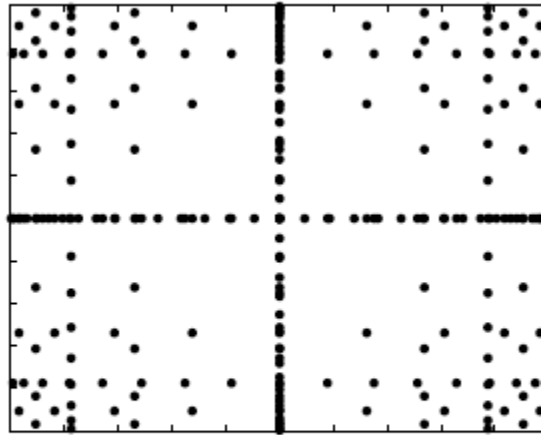


Sparse Grids

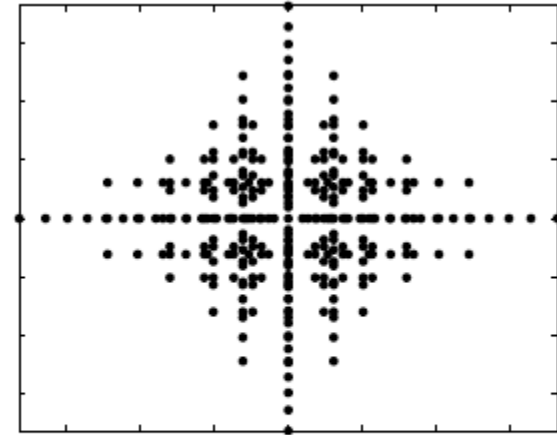
Choice of collocation points



(a) Clenshaw-Curtis



(b) Gauss



(c) Gauss-Hermite

Known problems: Oscillations of fitting polynomials

Smolyak Algorithm

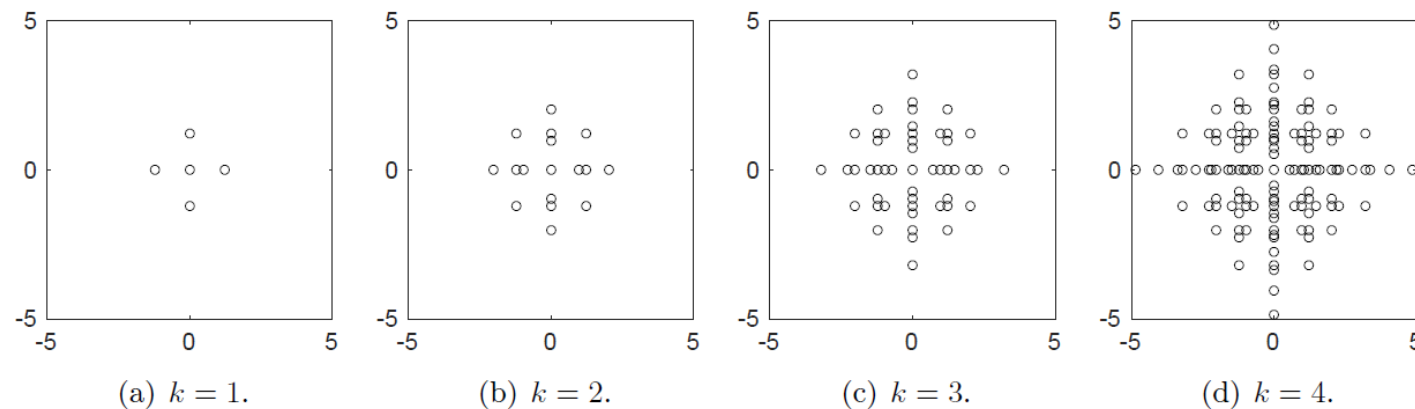
Interpolant

$$\mathcal{I}(u) = \sum_{k+1 \leq |i| \leq k+N} \eta(|i|) \cdot (\mathcal{U}^{i_1} \otimes \dots \otimes \mathcal{U}^{i_N})$$

where $|i| = i_1 + \dots + i_N$

with $\eta(|i|) = (-1)^{q-|i|} \cdot \binom{N-1}{q-|i|}$ with $q = k + N$

Estimate for number of points $n_{cp} \sim \frac{2^k}{k!} N^k$



Smolyak Algorithm

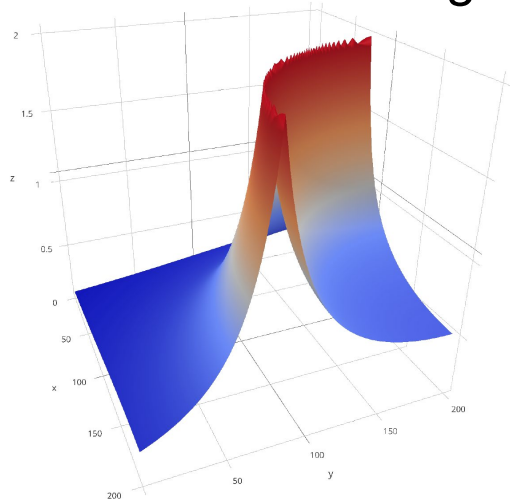
Statistics

$$\mathbb{E}\{u\} = \sum_{q-N+1 \leq |i| \leq q} \eta(|i|) \cdot \sum_{k=1}^{m_{i_1}} \cdots \sum_{k=N}^{m_{i_N}} u(X_{k_1}^{i_1}, \dots, X_{k_N}^{i_N}) \cdot \int_{\Gamma} (L_{k_1}^{i_1} \otimes \cdots \otimes L_{k_N}^{i_N}) f_X d\mathbf{X}$$

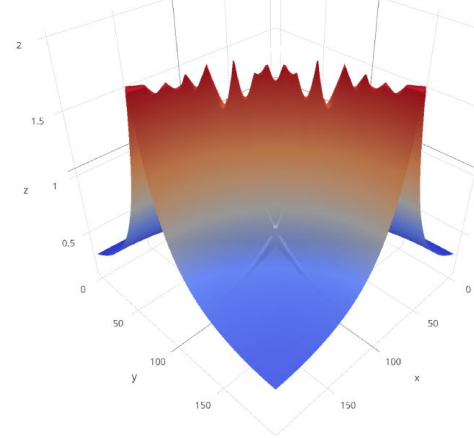
$$\text{Var}\{u\} = \sum_{q-N+1 \leq |i| \leq q} \eta(|i|) \cdot \sum_{k=1}^{m_{i_1}} \cdots \sum_{k=N}^{m_{i_N}} [u(X_{k_1}^{i_1}, \dots, X_{k_N}^{i_N}) - \mathbb{E}\{u\}]^2 \cdot \int_{\Gamma} (L_{k_1}^{i_1} \otimes \cdots \otimes L_{k_N}^{i_N}) f_X d\mathbf{X}$$

Adaptive Sparse Grids

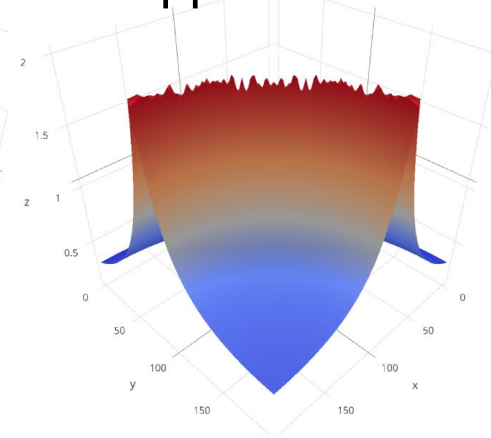
Known function with singularity



Clenshaw-Curtis Grid Approximation

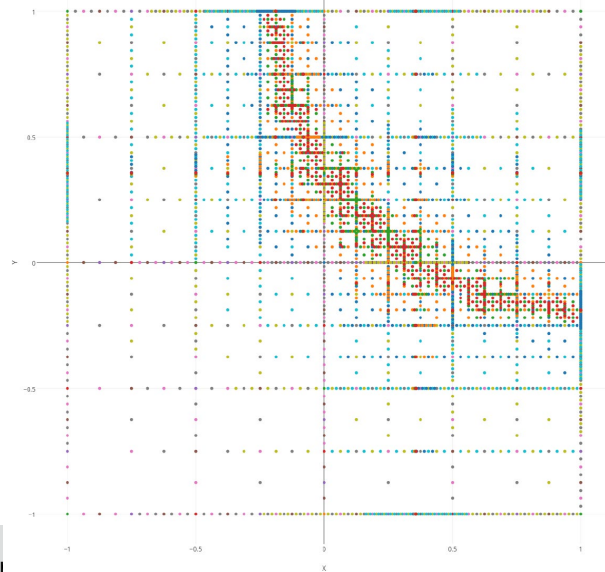


3329 points

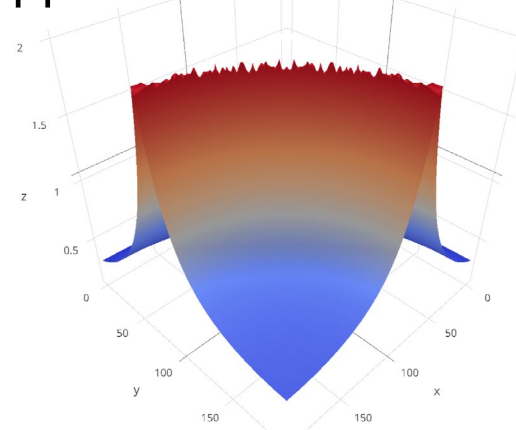


32769 Points

Adaptive Multi-Level Grid



Approximation with 3488 points



Perturbation Method

Taylor Series expansion

$$\mathbf{K}(\theta) = \mathbf{K}_0 + \sum_{i=1}^N \mathbf{K}_i^I \mathbf{X}_i + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \mathbf{K}_{ij}^{II} \mathbf{X}_i \mathbf{X}_j + \dots$$

$$\mathbf{U}(\theta) = \mathbf{U}_0 + \sum_{i=1}^N \mathbf{U}_i^I \mathbf{X}_i + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \mathbf{U}_{ij}^{II} \mathbf{X}_i \mathbf{X}_j + \dots$$

$$\mathbf{F}(\theta) = \mathbf{F}_0 + \sum_{i=1}^N \mathbf{F}_i^I \mathbf{X}_i + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \mathbf{F}_{ij}^{II} \mathbf{X}_i \mathbf{X}_j + \dots$$

zero-mean random vector $\mathbf{X} = [X_1, X_2, \dots, X_N]$

derivatives

$$\mathbf{K}_i^I = \left. \frac{\partial \mathbf{K}(\theta)}{\partial X_i} \right|_{\mathbf{X}=0}$$

$$\mathbf{K}_{ij}^{II} = \left. \frac{\partial^2 \mathbf{K}(\theta)}{\partial X_i \partial X_j} \right|_{\mathbf{X}=0}$$

Perturbation Method

Solution Scheme

$$\begin{aligned}\mathbf{U}_0 &= \mathbf{K}_0^{-1} \mathbf{F}_0 \\ \mathbf{U}_i^I &= \mathbf{K}_0^{-1} (\mathbf{F}_i^I - \mathbf{K}_i^I \mathbf{U}_0) \\ \mathbf{U}_{ij}^{II} &= \mathbf{K}_0^{-1} (\mathbf{F}_{ij}^{II} - \mathbf{K}_i^I \mathbf{U}_j^I - \mathbf{K}_j^I \mathbf{U}_i^I - \mathbf{K}_{ij}^{II} \mathbf{U}_0)\end{aligned}$$

Expected value

$$\mathbb{E}\{\mathbf{U}\} \approx \mathbf{U}_0 + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \mathbf{U}_{ij}^{II} \text{Cov}(X_i, X_j)$$

Limited to problems with small input variation

Neumann Expansion Method

In combination with MCS (avoiding multiple factorization)

$$\mathbf{K}(\theta) = \mathbf{K}_0 + \Delta\mathbf{K}$$

$$[\mathbf{K}_0 + \Delta\mathbf{K}] \mathbf{U} = \mathbf{F}$$

$$\mathbf{U}_0 = \mathbf{K}_0^{-1} \mathbf{F}$$

Neumann expansion of the stiffness matrix yields [Yamazaki, 1988]

$$\mathbf{K}^{-1}(\theta) = [\mathbf{K}_0 + \Delta\mathbf{K}]^{-1} \approx (\mathbf{I} - \mathbf{P} + \mathbf{P}^2 - \mathbf{P}^3 \dots) \mathbf{K}_0^{-1}$$

with $\mathbf{P} = \mathbf{K}_0^{-1} \Delta\mathbf{K}(\theta)$

solution $\mathbf{U} \approx \mathbf{U}_0 + \sum_n (-1)^n \mathbf{P}^n \mathbf{U}_0$

statistics

$$\mathbb{E}\{\mathbf{U}\} = \mathbf{U}_0$$

$$\text{Cov}(\mathbf{U}, \mathbf{U}) = \mathbb{E}\{\mathbf{P} \mathbf{U}_0 \mathbf{U}_0^T \mathbf{U}^T\}$$

Polynomial Chaos Expansion

Model $\mathcal{M}(\mathbf{X})$, $\mathbf{X} = [X_i]_{i=1}^M$

PC-approximation $\mathcal{M}(\mathbf{X}(\theta)) = \sum_{i=0}^{\infty} \bar{\mathcal{M}}_i \bar{\Phi}_i(\theta) \approx \sum_{i=0}^{P-1} \bar{\mathcal{M}}_i \bar{\Phi}_i(\theta)$

Orthogonal polynomials $\bar{\Phi}_i \in \mathcal{L}^2(\Omega, \mathcal{F}, \mathcal{P})$

PC-representation of a random variable

$$X(\theta) = \sum_{i=1}^{\infty} \hat{X}_i \Psi_i(\xi(\theta)) \approx \sum_{i=1}^P \hat{X}_i \Psi_i(\xi(\theta))$$

Number of polynomials (series of binominal coefficients)

$$P = \sum_{i=1}^p \binom{M+i-1}{i} \quad \begin{array}{l} p : \text{polynomial order} \\ M : \text{dimension of the stochastic space} \end{array}$$

Advantage: No basic assumptions with regard to the correlation structure of the random field!

Disadvantage: Special arithmetics needed

e.g. multiplication $Z = X \hat{\otimes} Y$

$$Z = \sum_{k=1}^P \hat{Z}_k \Psi_k = \sum_{i=1}^P \hat{X}_i \Psi_i \hat{\otimes} \sum_{j=1}^P \hat{Y}_j \Psi_j$$

yields

$$\hat{Z}_k = \sum_{i=1}^P \sum_{j=1}^P C_{ijk} \hat{X}_i \hat{\otimes} \hat{Y}_j$$

with the coefficients

$$C_{ijk} = \frac{\langle \Psi_i \Psi_j \Psi_k \rangle}{\Psi_k^2} \quad (\forall (i, j, k) \in (1, \dots, P)^3)$$

Hierarchical Hermite Polynomials

j	p , order of the homogeneous chaos	j -th polynomial chaos ψ_j	$\langle \psi_j^2 \rangle$
0	$p = 0$	1	1
1	$p = 1$	ξ_1	1
2		ξ_2	1
3	$p = 2$	$\xi_1^2 - 1$	2
4		$\xi_1 \xi_2$	1
5		$\xi_2^2 - 1$	2
6	$p = 3$	$\xi_1^3 - 3\xi_1$	6
7		$\xi_1^2 \xi_2 - \xi_2$	2
8		$\xi_1 \xi_2^2 - \xi_1$	2
9		$\xi_2^3 - 3\xi_2$	6
10	$p = 4$	$\xi_1^4 - 6\xi_1^2 + 3$	24
11		$\xi_1^3 \xi_2 - 3\xi_1 \xi_2$	6
12		$\xi_1^2 \xi_2^2 - \xi_1^2 - \xi_2^2 + 1$	4
13		$\xi_1 \xi_2^3 - 3\xi_1 \xi_2$	6
14		$\xi_2^4 - 6\xi_2^2 + 3$	24

Generalized Polynomial Chaos

General: Family of orthogonal hypergeometric functions!

	PDF	Orthogonal polynomials	Support range
Continuous	GAUSSIAN	HERMITE	$(-\infty, \infty)$
	Gamma	LAGUERRE	$[0, \infty)$
	Uniform	Legendre	$[a, b]$
	Beta	Jacobi	$[a, b]$
Discrete	Possion	CHARLIER	$\{0, 1, 2, \dots\}$
	Binomial	KRAWTCHOUK	$\{0, 1, 2, \dots N\}$
	negative Binomial	MEIXNER	$\{0, 1, 2, \dots\}$
	Hypergeometric	HAHN	$\{0, 1, 2, \dots N\}$

Number P of polynomials

M	$p = 1$	$p = 2$	$p = 3$	$p = 4$	$p = 5$	$p = 6$
2	3	6	10	15	21	28
4	5	15	35	70	126	210
6	7	28	84	210	462	924

Generalized Polynomial Chaos (gPC) [Xiu, 2014]

Spectral Stochastic Finite Element Method

Direct processing of the approximation of the stochastic variables into the Galerkin based Finite Element Procedure.

$$\mathbf{K}(\theta)\mathbf{u}(\theta) = \mathbf{r}(\theta)$$

e.g. with the Polynomial Chaos expansion of the random variables

$$\mathbf{K}(\theta) = \sum_{i=1}^P \hat{\mathbf{K}}_i \Psi_i(\boldsymbol{\xi})$$

results in

$$\sum_{i=1}^P \hat{\mathbf{K}}_i \Psi_i(\boldsymbol{\xi}) \sum_{j=1}^P \hat{\mathbf{u}}_j \Psi_j(\boldsymbol{\xi}) = \sum_{k=1}^P \hat{\mathbf{r}}_k \Psi_k(\boldsymbol{\xi})$$

Galerkin orthogonality for Gaussian PC (Hermitian approximation) yields

$$\sum_{i=1}^P \sum_{j=1}^P C_{ijk} \hat{\mathbf{K}}_i \hat{\mathbf{u}}_j = \hat{\mathbf{r}}_k$$

SSFEM

Resulting FE-system:

$$\begin{bmatrix} \mathbf{K}_{11} & \cdots & \mathbf{K}_{1P} \\ \vdots & \ddots & \vdots \\ \mathbf{K}_{P1} & \cdots & \mathbf{K}_{PP} \end{bmatrix} \begin{bmatrix} \mathbf{u}_1 \\ \vdots \\ \mathbf{u}_P \end{bmatrix} = \begin{bmatrix} \mathbf{r}_1 \\ \vdots \\ \mathbf{r}_P \end{bmatrix}$$

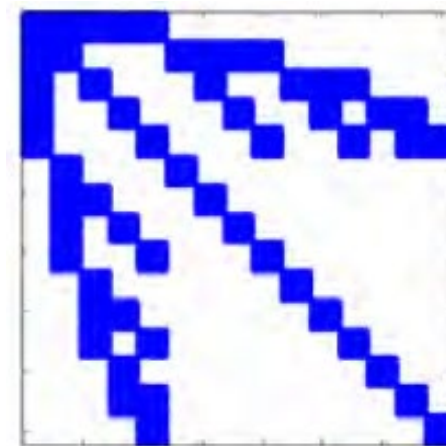
with

$$\mathbf{K}_{ij} \sum_{i=1}^P C_{ijk} \hat{\mathbf{K}}_i$$

Gauss-Seidel type solution scheme

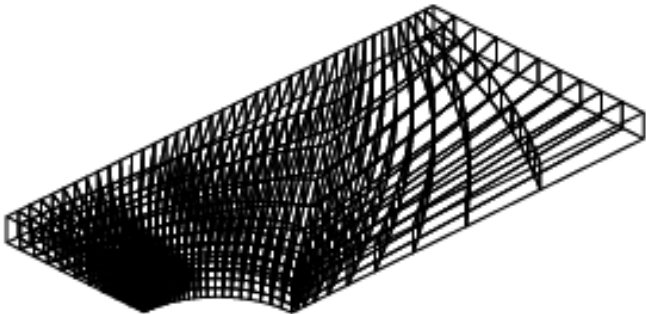
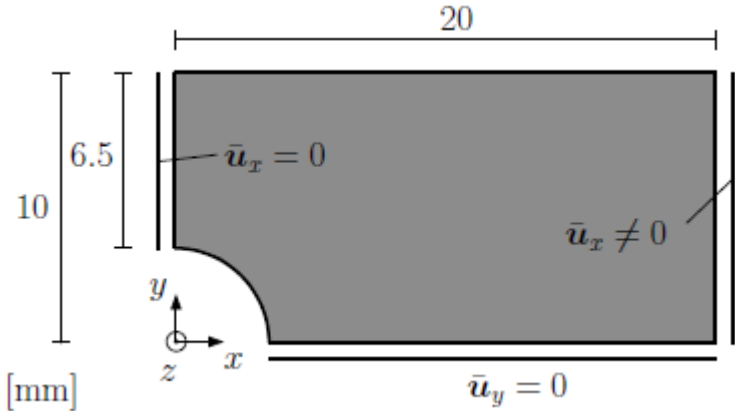
$$\mathbf{K}_{ii} \mathbf{u}_i = \mathbf{r}_i - \sum_{1 \leq j < i} \mathbf{K}_{ij} \mathbf{u}_j$$

typical sparsity pattern

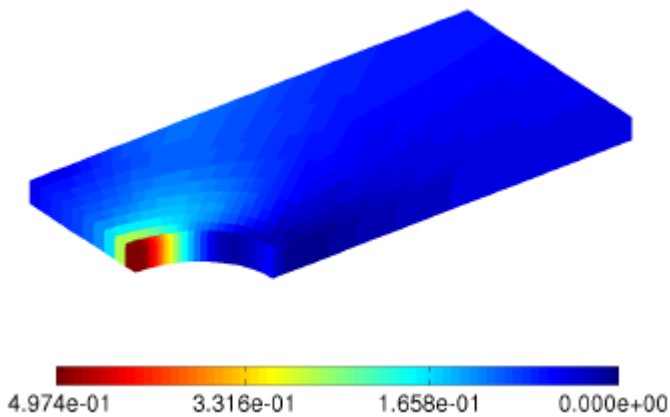


Comparison of Different Approaches

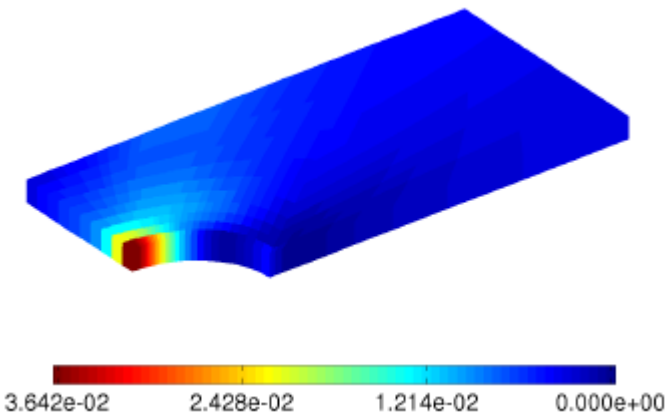
Disc with hole



Expectation (D)



Standarddeviation



Comparison of Different Approaches

Comparison of computational effort

method	#H	CPU/sec
MC	500	18650
SSFEM	1	34900
Collocation-F	64	2570
Collocation-S	65	2700

Non-Linear Models

e.g. ductile damage model with the strain energy density function

$$\psi(\boldsymbol{\varepsilon}, D, a) = \frac{1}{2\rho} \boldsymbol{\varepsilon}^{(e)} \cdot \cdot (1 - D) \mathbb{C} \cdot \cdot \boldsymbol{\varepsilon}^{(e)} + \psi^{(pl)}(a)$$

Yield function

$$f = \frac{\sqrt{2J_2(\text{dev}\boldsymbol{\sigma})}}{1 - D} - \sigma_y(\sigma_{y0}, \alpha)$$

Evolution rules

$$\dot{\boldsymbol{\varepsilon}}^{(p)} = \dot{\lambda} \boldsymbol{n} \quad \text{with} \quad \boldsymbol{n} = \sqrt{\frac{2}{3}} \frac{\text{dev}(\boldsymbol{\sigma})}{(1 - D) \|\text{dev}(\boldsymbol{\sigma})\|}$$

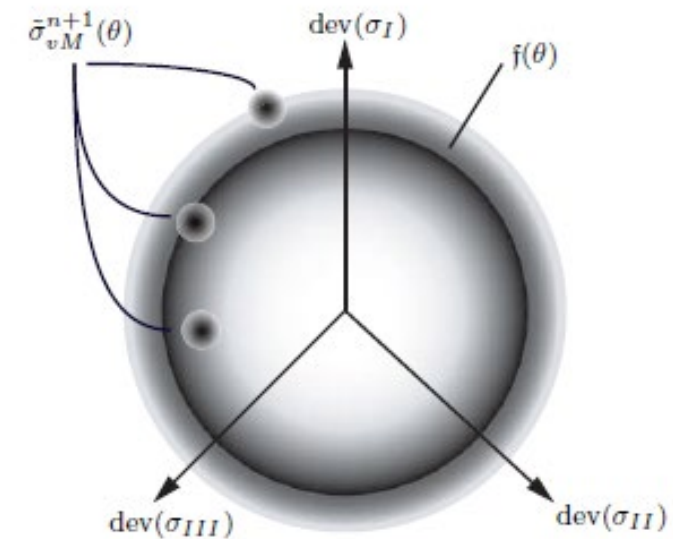
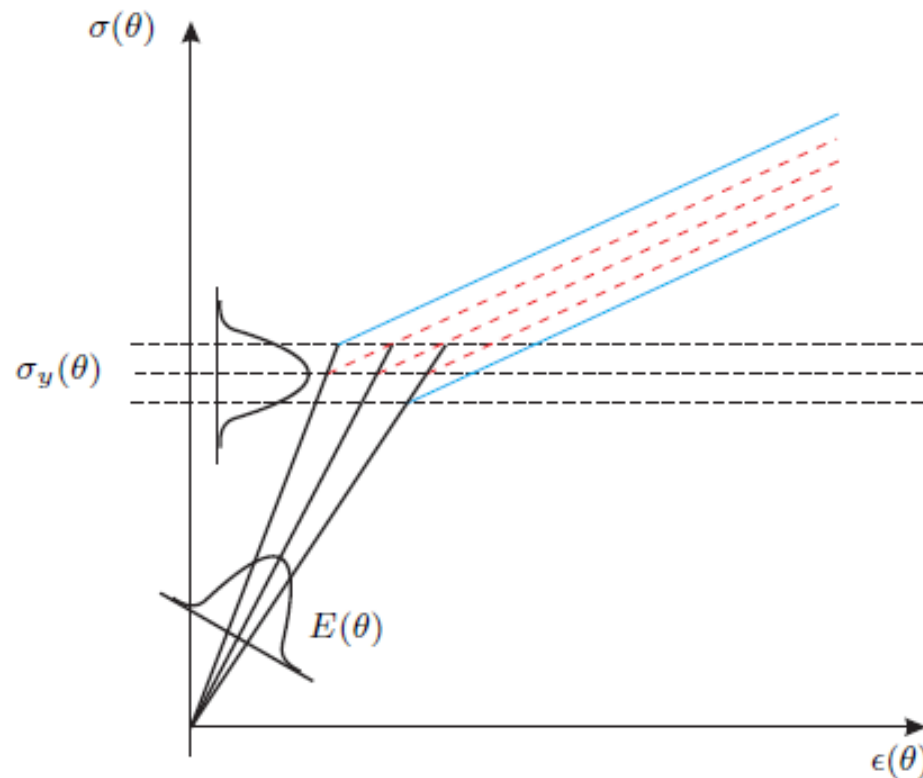
$$\dot{\alpha} = \dot{\lambda}$$

$$\dot{D} = \frac{\dot{\lambda}}{1 - D} \left(-\frac{Y}{r} \right)^s \quad \text{with} \quad Y = -\frac{\sigma_y^2}{6\mu} - \frac{p^2}{2\kappa}$$

Non-Linear Models

Remark on Radial Return Mapping Scheme

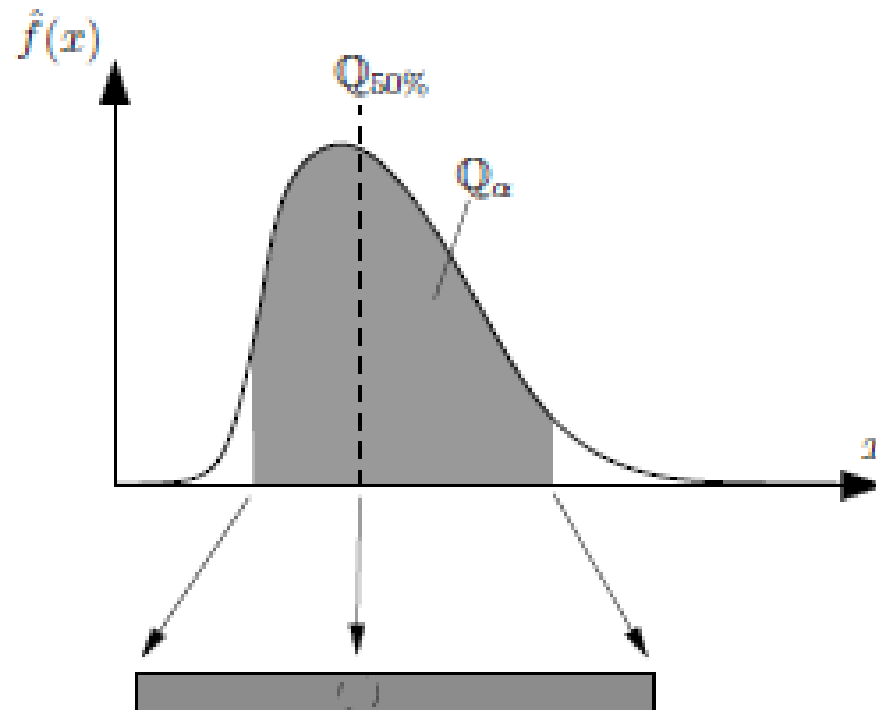
Yield surface is random now!



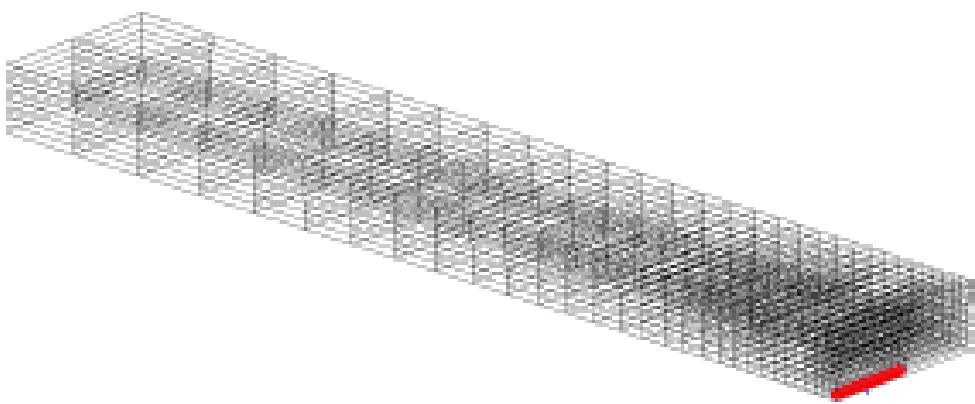
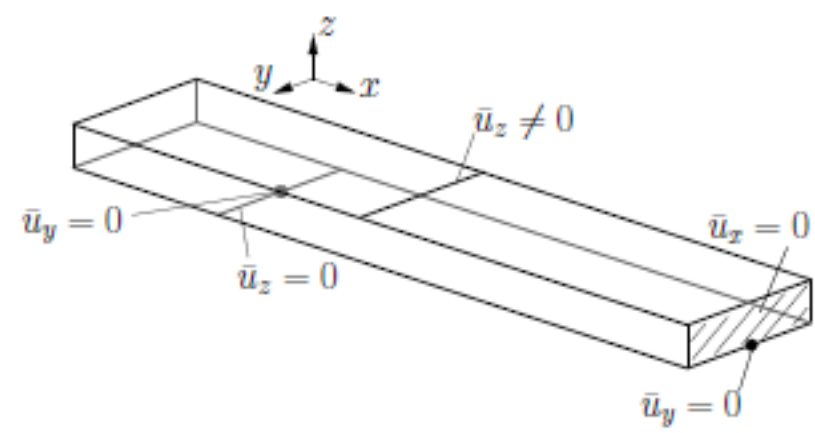
Remarks on Stochastic Return Mapping

Common approach: Usage of mean values!

Alternative approach: Usage of Quantils



Numerical Example

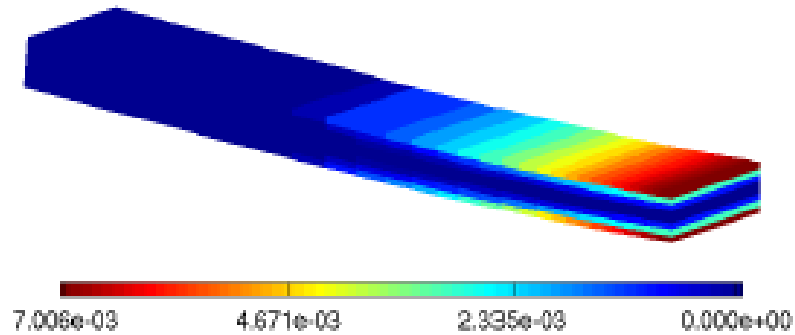


Parameter	Verteilung	$\langle \bullet \rangle$	$SD(\bullet)$
E	lognormal	210000 N/mm ²	10500 N/mm ²
σ_{y0}	lognormal	235.0 N/mm ²	11.75 N/mm ²
s	lognormal	4.0	0.2
r	lognormal	0.2	0.01
ν	-	0.3	-
$\sigma_{y\infty}$	-	180 N/mm ²	-
γ	-	100	-

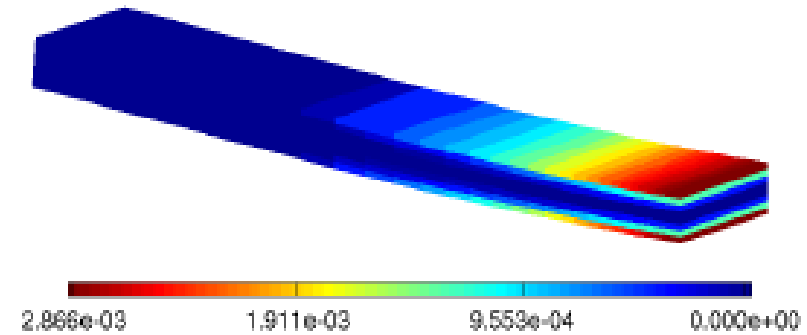
4-Point Bending

mean of damage variable

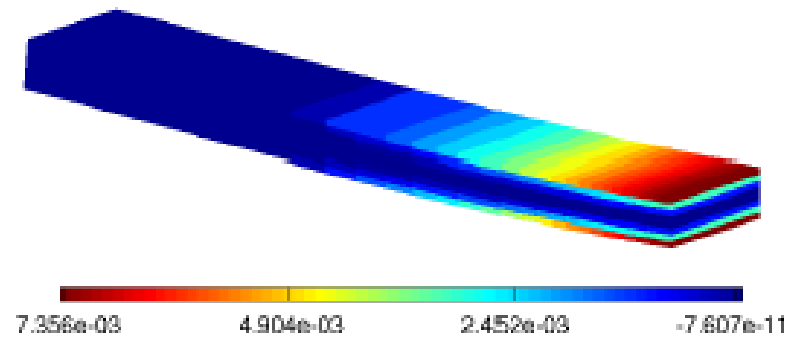
standard deviation of damage variable



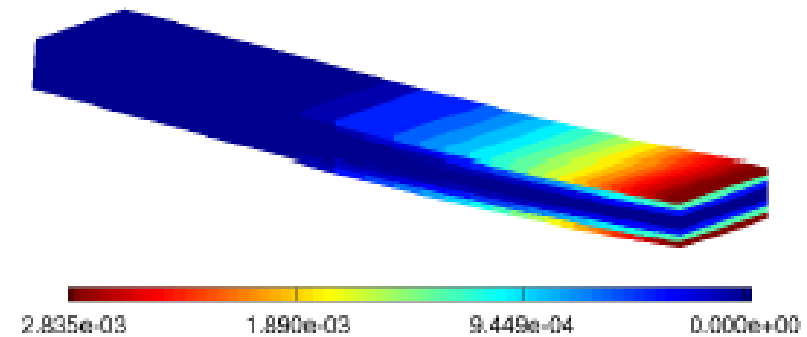
(a) MC



(e) MC



(b) KS (PDF)

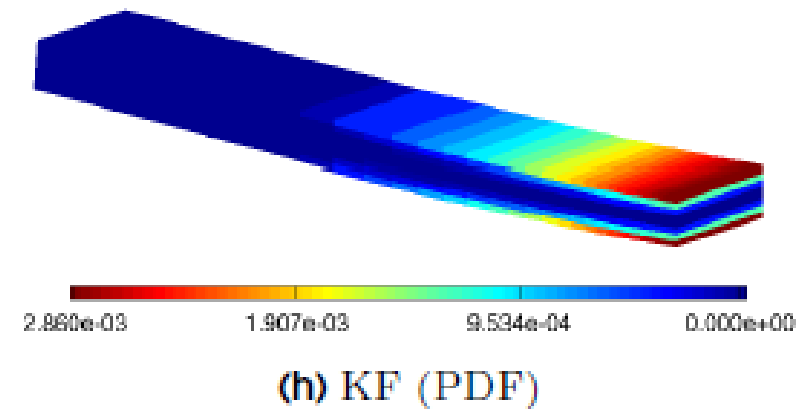
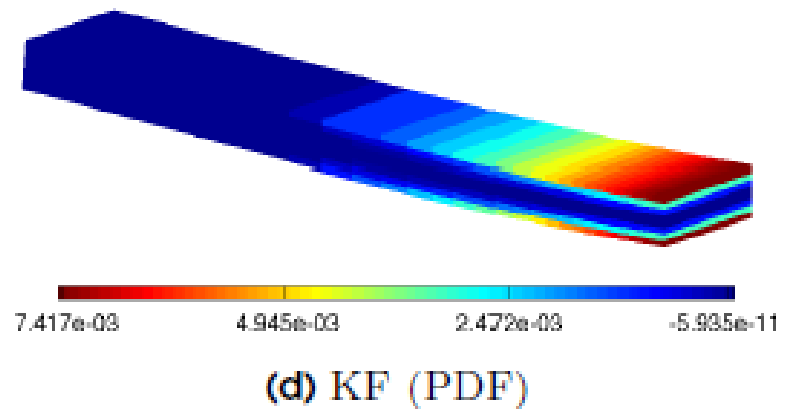
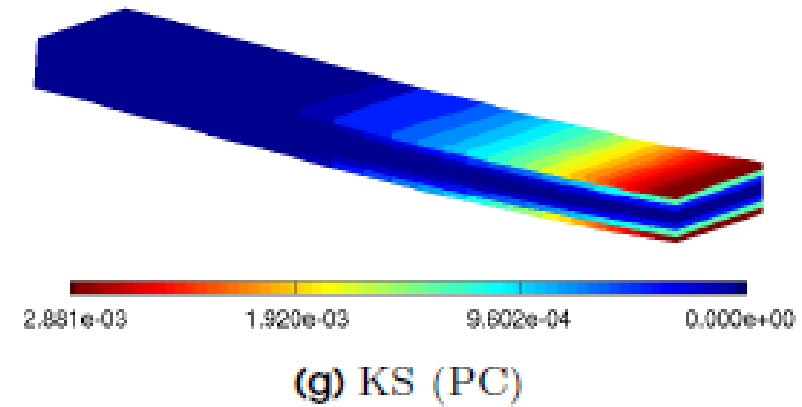
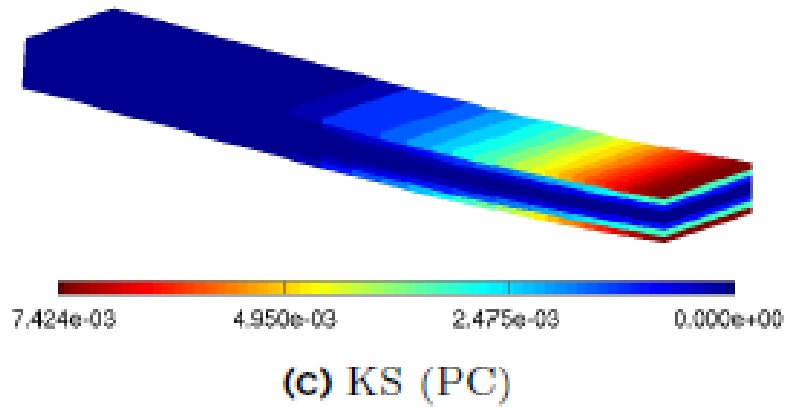


(f) KS (PDF)

4-Point Bending

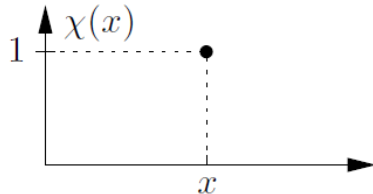
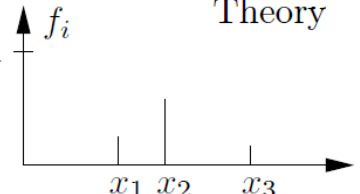
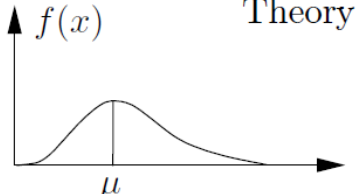
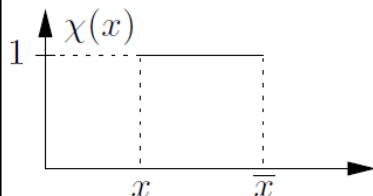
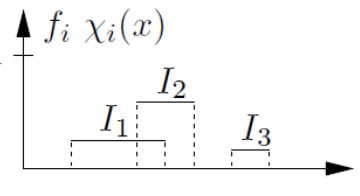
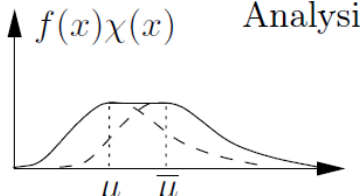
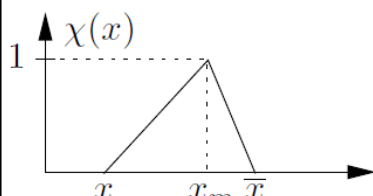
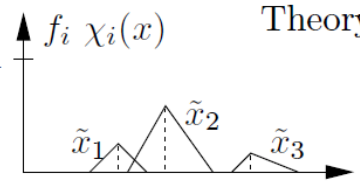
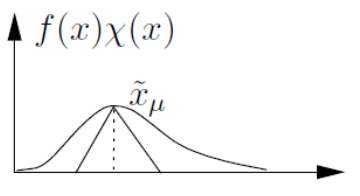
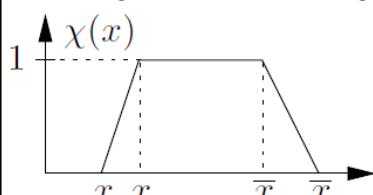
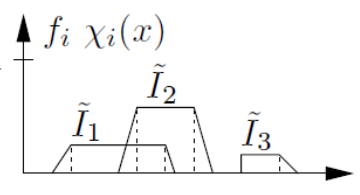
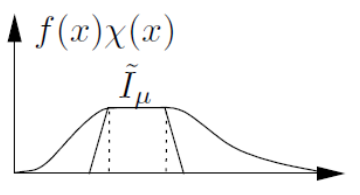
mean of damage variable

standard deviation of damage variable



Classification of Uncertainties

- **aleatory**: inherent randomness of physical properties, can not be reduced by additional information -> probabilistic approaches, e.g.
 - Monte Carlo Methods
 - Advanced sampling techniques
 - Collocation methods
 - Spectral Stochastic Finite Element Methods
- **epistemic**: lack of knowledge, limitations in building the mathematical model of a physical system; can be reduced by additional information -> non-probabilistic (possibilistic) approaches, e.g.
 - Bayes updating
 - Evidence theory
 - Interval analysis
 - Probability boxes
 - Fuzzy arithmetic

		Uncertainty	Probability Theory	
Imprecision	↓	Parameter	Finite	Infinite
Classical Set Theory	Number	(a) Deterministic 	(b) Discrete Probability Theory 	(c) Continuous Probability Theory 
	Interval	(d) Interval Analysis 	(e) Evidence Theory 	(f) Probability Bounds Analysis 
Fuzzy Set Theory	Fuzzy Number	(g) Fuzzy Analysis 	(h) Fuzzy Probability Theory 	(i) 
	Fuzzy Interval	(j) Fuzzy Interval Analysis 	(k) 	(l) 
Height		$h = 1$	$\sum h_i = 1$	$\int h(x) dx = 1$

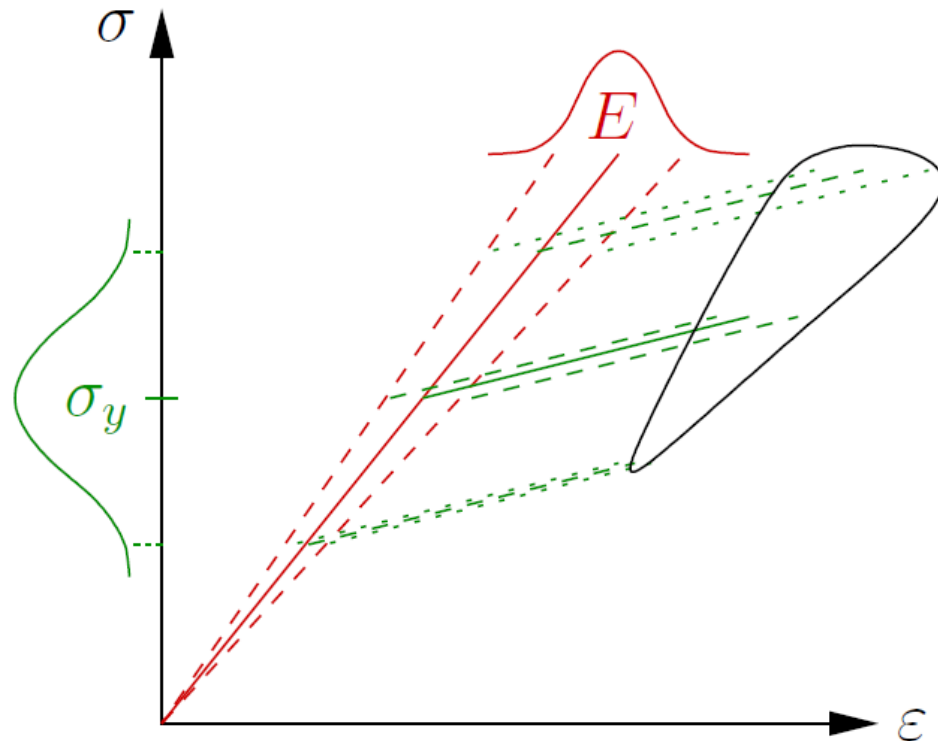
e.g. Random Fields

Uncertainties within the random field

- *Aleatory* (randomness): stochastic parameters
 - Mean value μ
 - Variance σ^2
 - Modelled by Karhunen-Loève expansion
 - *Epistemic* (lack of knowledge): spatial parameters
 - Correlation length l_c
 - Assumed to be interval-valued
- Need for approaches to tackle mixed types of uncertainties

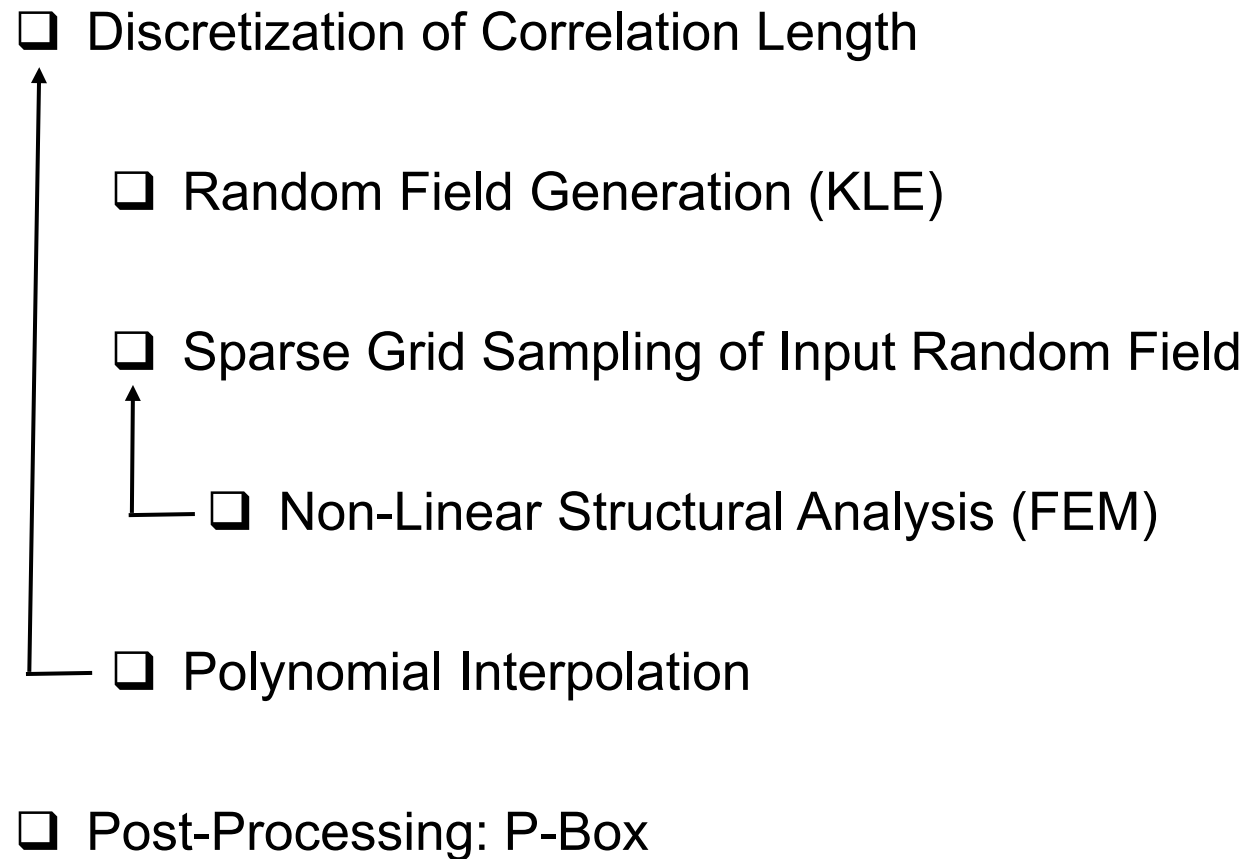
Computational Techniques

Challenges in non-linear finite element simulations



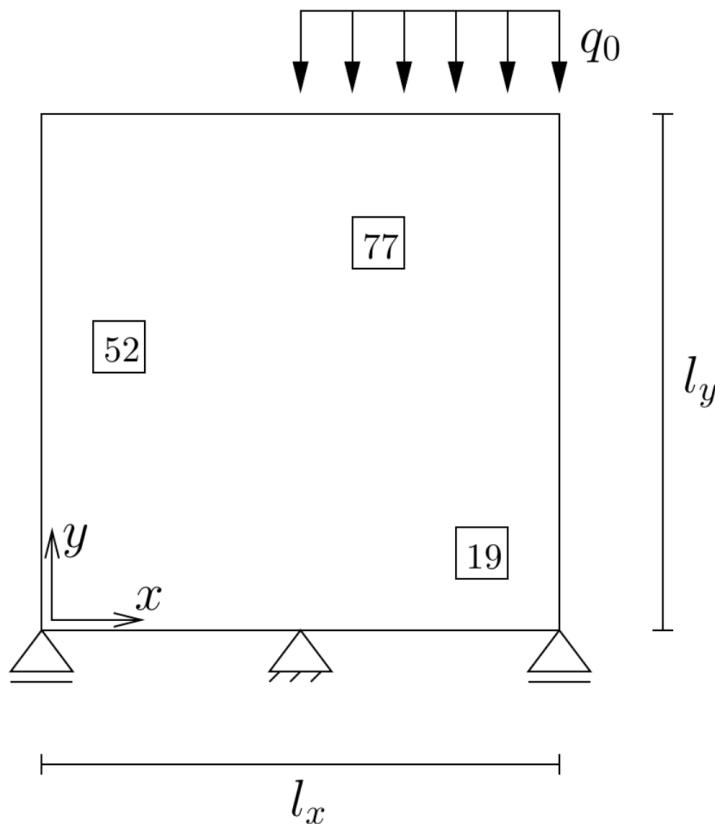
- ☐ Plenty of samples to be computed
- ☐ Additional outer loop for discretised P-Box
- ☐ Efficient sampling techniques
- ☐ Model reduction techniques

Nested Collocation Scheme



Numerical Example

Square plate of dimension $l_x = l_y = 1$ m discretized with 100 linear finite elements and loaded by $q_0 = -500 \text{ kN} \cdot \text{mm}^{-1}$



Young's Modulus $E = 210 \text{ GPa}$

Poisson's Ratio $\nu = 0.28$

Linear kinematic hardening $H_{kin} = 1.5 \text{ GPa}$

Gaussian Yield stress distribution

$$E\{\sigma_y\} = 240 \text{ MPa}$$

$$\text{Std}\{\sigma_y\} = 12 \text{ MPa}$$

Interval valued correlation length

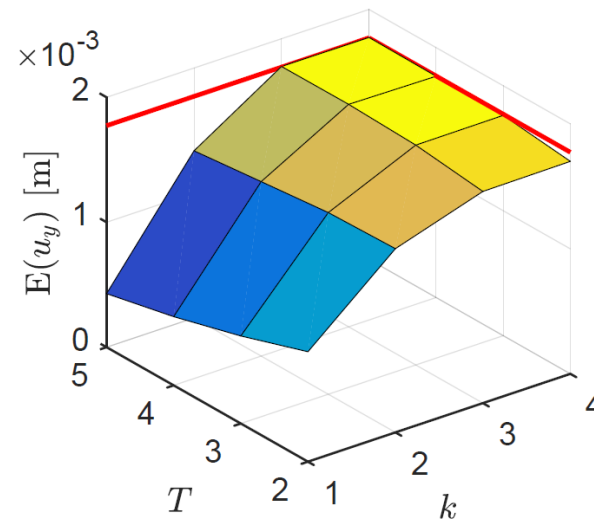
$$l_c \in [0.5, 5.0] \text{ m}$$

Numerical Example

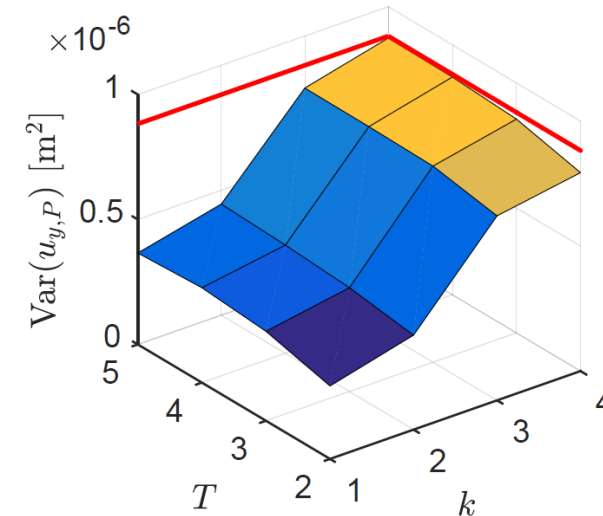
Convergence Studies

Lower bound: $l_c = 0.5\text{m}$

Reference (red line): Monte Carlo ($n^{MC} = 10^5$ samples)



Expected value

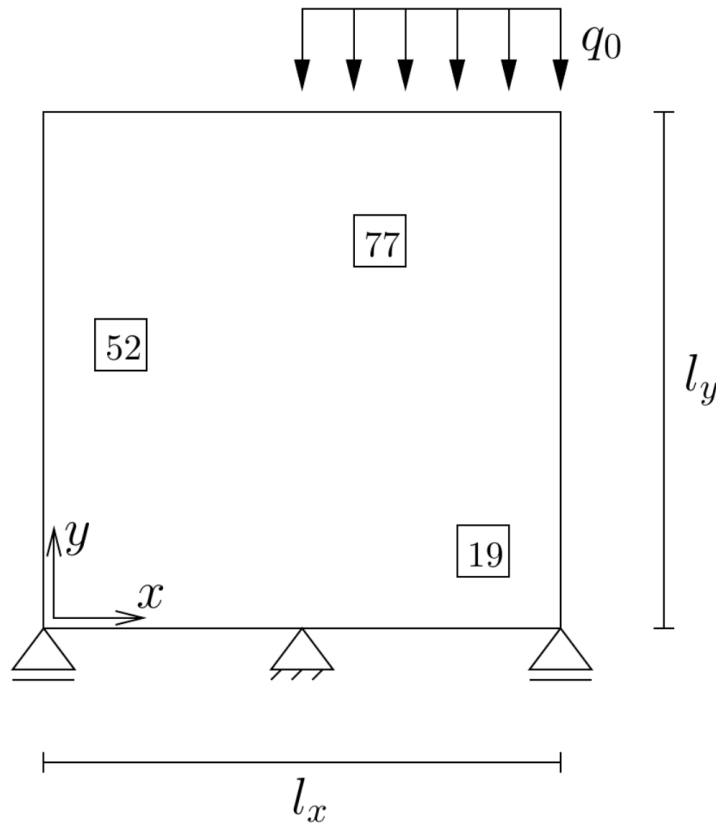


Variance

$k = 4, T = 5 : n^{SC} = 1471$ collocation points

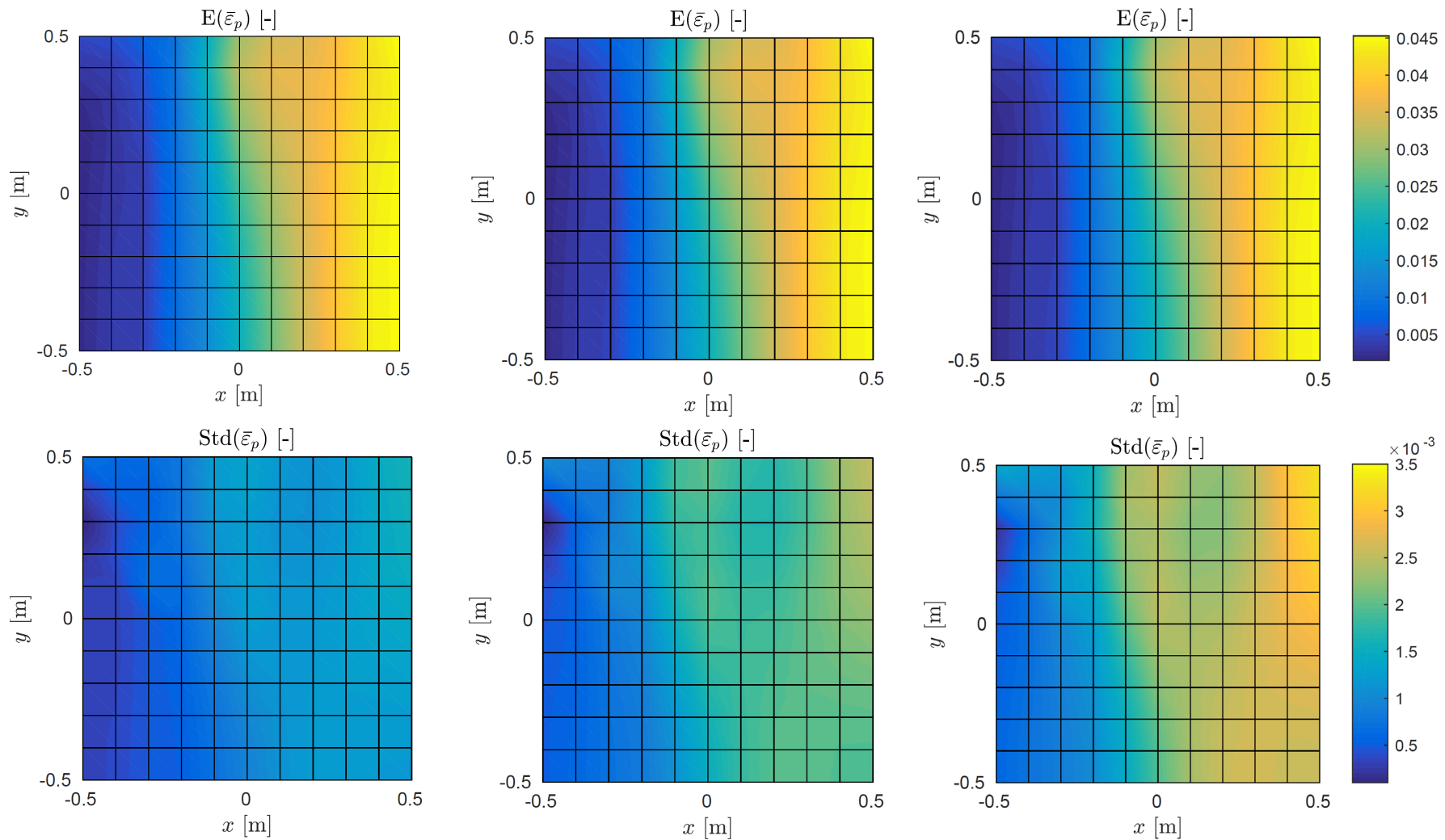
$\rightarrow \text{err} < 1\%, \quad t^{SC} \simeq 0.13 t^{MC}$

Numerical Example



- ❑ Random Field Approximation truncated after $T = 5$ terms.
- ❑ Gaussian Hermite Sparse Grid with Smolyak level $k = 4$.
- ❑ Deterministic Non-linear Finite Element Simulation in 20 load steps.

Numerical Example

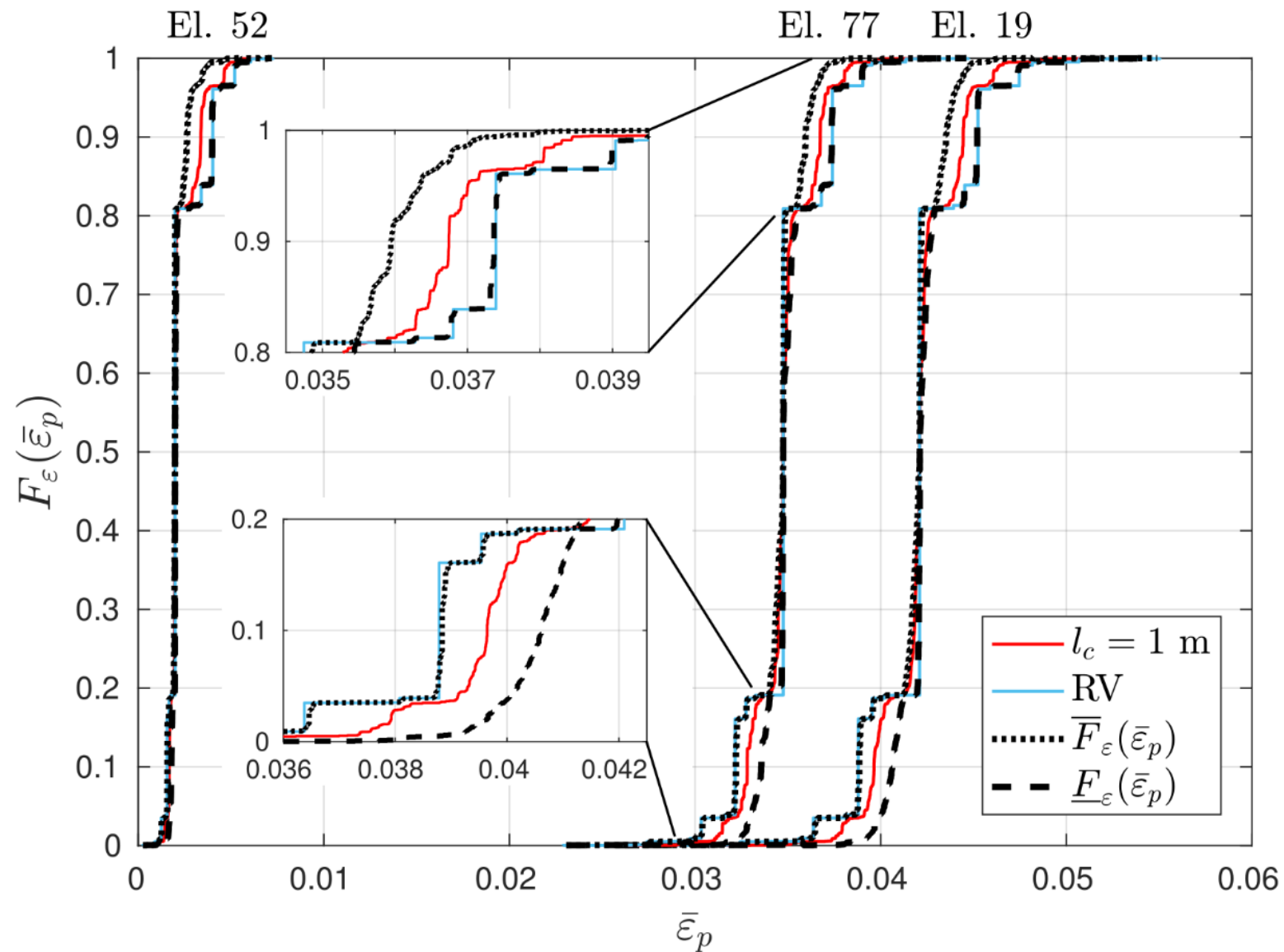


$l_c = 0.5$ m

$l_c = 1.0$ m

$l_c = 5.0$ m

Numerical Example



Traditional Approaches

Stochastic Processes

We assume Markov properties (transition probability)

$$f_{1|n-1}(\mathbf{x}_n, t_n | \mathbf{x}_1, t_1; \cdots \mathbf{x}_{n-1}, t_{n-1}) = f_{1|1}(\mathbf{x}_n, t_n | \mathbf{x}_{n-1}, t_{n-1})$$

which for multi-dimensional problems can be written as products of transition probability densities,

$$f_n(\mathbf{x}_1, t_1; \cdots \mathbf{x}_n, t_n) = f_1(\mathbf{x}_1, t_1) f_{1|1}(\mathbf{x}_2, t_2 | \mathbf{x}_1, t_1) \cdots f(\mathbf{x}_n, t_n | \mathbf{x}_{n-1}, t_{n-1})$$

Integration over an intermediate realization yields

$$f_{1|1}(\mathbf{x}_k, t_k | \mathbf{x}_i, t_i) = \int_{\mathbb{R}} f_{1|1}(\mathbf{x}_k, t_k | \mathbf{x}_j, t_j) f_{1|1}(\mathbf{x}_j, t_j | \mathbf{x}_i, t_i) \, d\mathbf{x}_j$$

which is known as *Chapman-Kolmogorov* equation.

Chapman-Kolmogorov Equation

Definition after Gardiner, 2009

$$\begin{aligned} \frac{\partial f(\mathbf{x}_c, t_c | \mathbf{x}_a, t_a)}{\partial t} = & - \sum_i \frac{\partial}{\partial x_{c,i}} [A_i(\mathbf{x}_c, t_c) f(\mathbf{x}_c, t_c | \mathbf{x}_a, t_a)] \\ & + \frac{1}{2} \sum_{i,j} \frac{\partial^2}{\partial x_{c,i} \partial x_{c,j}} [B_{ij} f(\mathbf{x}_c, t_c | \mathbf{x}_a, t_a)] \\ & + \int_{\mathbb{R}}^n [W(\mathbf{x}_c | \mathbf{x}_a) f(\mathbf{x}_b, t_b | \mathbf{x}_a, t_a) \\ & - W(\mathbf{x}_b | \mathbf{x}_c) f(\mathbf{x}_c, t_c | \mathbf{x}_a, t_a)] d\mathbf{x}_b \end{aligned}$$

Where $\mathbf{A}$ and $\mathbf{B}$ are the drift vector and diffusion matrix, respectively.

The second part describes the jump transition probability.

Master Eqn. / Fokker-Planck Eqn.

Definition after Gardiner, 2009

$$\frac{\partial f(\mathbf{x}_c, t_c | \mathbf{x}_a, t_a)}{\partial t} = \int_{\mathbb{R}}^n [W(\mathbf{x}_c | \mathbf{x}_a) f(\mathbf{x}_b, t_b | \mathbf{x}_a, t_a) - W(\mathbf{x}_b | \mathbf{x}_c) f(\mathbf{x}_c, t_c | \mathbf{x}_a, t_a)] d\mathbf{x}_b$$

Fokker-Planck Equation

$$\frac{\partial f}{\partial t} = - \sum_i^r \frac{\partial}{\partial x_i} [A_i(\mathbf{x}) f] + \frac{1}{2} \sum_{i,j}^r \frac{\partial^2}{\partial x_i \partial x_j} [B_{ij}(\mathbf{x}) f]$$

Origin in physics: motion of powder in liquids.

